

Reconstructing frustrated many-body signs from amplitudes with fixed-node and Krylov updates

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The sign structure of a frustrated many-body wave function is difficult to sample and difficult to learn, even when its positive amplitude is already accurate. We study whether the two tasks can be separated in the square-lattice spin-1/2 J_1 - J_2 Heisenberg model. Writing a real wave function as $\psi(x) = s(x)a(x)$ with $a(x) > 0$, we update the sign from a single Krylov vector, while fixed-node projection supplies information about the amplitude.

For a given amplitude, the sign of $(T - H)\psi$ is determined by a scalar threshold of the local energy. On a 4×4 exact-diagonalization benchmark at $J_2/J_1 = 0.5$, this one-step rule reduces the exact wrong-sign probability mass from 1.27×10^{-2} to 3.41×10^{-4} and the fixed-amplitude energy error from 1.16×10^{-2} to 5.51×10^{-4} per site. On 6×6 , with the positive amplitude held fixed, the reconstructed signs reproduce the energy of a trained neural quantum-state sign structure within the sampling uncertainty. With those signs, fixed-node projection remains stable on 8×8 and yields an energy in the same range as high-accuracy variational benchmarks.

The converse step can also be closed. On 4×4 , an exact fixed-node amplitude followed by the Krylov sign update recovers the exact ground-state signs from a nonfrustrated starting state. For sampled fixed-node walkers, maximum likelihood has an exact target: $a_{\text{new}} \propto \sqrt{a\phi_{\text{FN}}}$. At 8×8 , ordinary Euclidean likelihood gradients from independent walker populations are strongly inconsistent, whereas a stochastic-reconfiguration direction improves a held-out population. These results identify one-step sign reconstruction as the simple part of the loop and the conversion of fixed-node walkers into a transferable callable amplitude as the remaining scaling problem.

I. INTRODUCTION

The sign problem is one of the central obstacles to Monte Carlo calculations of frustrated quantum matter. In a basis where off-diagonal matrix elements cannot all be made nonpositive, positive and negative contributions cancel, and the cost of direct sampling can grow exponentially [1, 2]. The square-lattice J_1 - J_2 Heisenberg antiferromagnet is a standard example.

At $J_2 = 0$ the Marshall sign rule fixes the ground-state sign structure on the bipartite square lattice [3]. Around $J_2/J_1 = 1/2$, frustration invalidates that simple rule and simultaneously makes the model a demanding benchmark for modern variational methods, including neural quantum states [4, 5].

A common response is to improve the entire wave function at once. Neural quantum states can represent both amplitudes and signs [6], while fixed-node methods avoid the sign problem by holding a trial sign structure fixed [7]. Here we ask a different question: can the sign and amplitude be improved by different numerical operations?

The answer suggested by our data is yes. Given a positive amplitude $a(x)$ and a current sign pattern $s(x)$, the sign of one Krylov vector $(T - H)as$ can be read off from a scalar local quantity. Conversely, once the sign is fixed, a lattice fixed-node projector can improve the positive amplitude without a sign problem. This gives the feedback structure shown in Fig. 1. The method is not an attempt to make the original Hamiltonian stoquastic. It instead alternates between a sign-free amplitude problem and an

Separate amplitude and sign updates

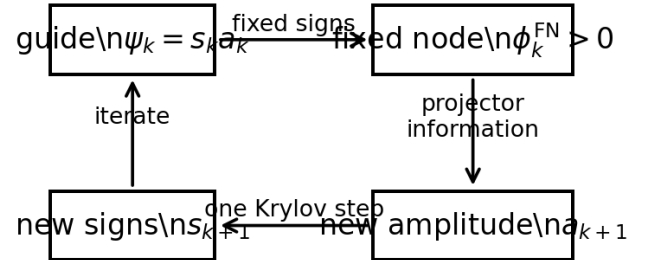


FIG. 1. Sign–amplitude feedback used throughout the paper. A guide $\psi_k = s_k a_k$ is projected with its signs held fixed. The resulting positive-amplitude information is converted into a new callable amplitude, and only then is a one-step Krylov update used to change the signs.

explicit sign-changing update.

Our tests are organized to isolate the two parts of this loop. Exact diagonalization first asks whether one Krylov step reconstructs the signs when the amplitude is known. A same-amplitude 6×6 comparison then asks whether the reconstructed signs have the same physical effect as the signs of a trained neural quantum state. An exact 4×4 feedback loop tests whether fixed-node amplitude improvement and sign reconstruction can self-correct when composed. Finally, 8×8 calculations iden-

tify the practical bottleneck: converting fixed-node walkers into a callable amplitude that generalizes beyond one walker population.

II. MODEL AND UPDATE RULES

We consider

$$H = J_1 \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + J_2 \sum_{\langle\langle ij \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_j, \quad (1)$$

on $L \times L$ periodic square lattices, with $J_1 = 1$. Calculations are performed in the $S^z = 0$ sector. Since Eq. (1) is real, the wave function can be written

$$\psi(x) = s(x)a(x), \quad s(x) = \pm 1, \quad a(x) > 0, \quad (2)$$

for spin configurations x .

When the exact ground state $\psi_0 = s_0 a_0$ is available, we quantify the sign error by

$$O_s = \left| \sum_x a_0(x)^2 s(x) s_0(x) \right|, \quad w_s = \frac{1 - O_s}{2}. \quad (3)$$

After fixing the irrelevant global sign, w_s is simply the exact ground-state probability carried by configurations with the wrong sign. For larger systems, where exact signs are unavailable, we instead compare energies at fixed amplitude or fixed-node projected energies.

A. One-step Krylov sign reconstruction

For a signed guide $\psi = sa$, define its local energy

$$r_s(x) = \frac{(H\psi)(x)}{\psi(x)}. \quad (4)$$

A single shifted Krylov vector satisfies

$$[(T - H)\psi](x) = [T - r_s(x)]\psi(x). \quad (5)$$

Therefore its sign is obtained without evaluating a new many-body amplitude:

$$s_K(x) = s(x) \operatorname{sgn}[T - r_s(x)]. \quad (6)$$

The amplitude factor $|T - r_s|a$ is discarded. We retain only the sign and restore or re-solve the amplitude before another sign update. This distinction is important: repeated Krylov multiplication is not the algorithm studied here.

The shift T acts as a one-dimensional decision threshold. In exact small-system tests we choose it by minimizing the fixed-amplitude variational energy after the sign update; this uses the Hamiltonian but no target sign labels. On sampled larger systems we determine T from the distribution of r_s itself with a robust two-cluster split. Exact or neural signs are used only afterward as diagnostics.

B. Fixed-node amplitude update

For a guide $\psi_G(x) = s(x)a(x)$, define the sign-violating neighbors

$$\mathcal{V}_x = \left\{ y \neq x : H_{xy} \frac{\psi_G(y)}{\psi_G(x)} > 0 \right\}. \quad (7)$$

The lattice fixed-node Hamiltonian removes every sign-violating hopping term and transfers its contribution to the diagonal:

$$(H_{\text{FN}})_{xy} = \begin{cases} H_{xy}, & y \notin \mathcal{V}_x, \\ 0, & y \in \mathcal{V}_x, \end{cases} \quad x \neq y, \quad (8)$$

$$(H_{\text{FN}})_{xx} = H_{xx} + \sum_{y \in \mathcal{V}_x} H_{xy} \frac{\psi_G(y)}{\psi_G(x)}. \quad (9)$$

After importance sampling with the guide, H_{FN} is sign free [7]. Its ground state has the imposed sign pattern $s(x)$ and a positive modulus that we denote by $\phi_{\text{FN}}(x)$.

On an exact small system one may simply feed that modulus back into Eq. (6). In a walker calculation, however, the projector does not return a table of $\phi_{\text{FN}}(x)$. With guide amplitude $a(x)$, the stationary mixed distribution is

$$f(x) \propto a(x)\phi_{\text{FN}}(x). \quad (10)$$

If a positive model is trained by maximum likelihood with Born probability $p_\theta(x) \propto a_\theta(x)^2$, the ideal maximum-likelihood target is therefore

$$a_{\text{new}}(x) \propto \sqrt{a(x)\phi_{\text{FN}}(x)}. \quad (11)$$

Equation (11) is the precise amplitude handoff that we later benchmark. It is a geometric-mean, or half, step toward the current fixed-node amplitude; because H_{FN} itself depends on the guide ratios, the target changes after every accepted update.

C. Numerical protocol

The exact 4×4 calculations use the full $S^z = 0$ Hilbert space of dimension 12870. For 6×6 and 8×8 we use a previously optimized translationally invariant complex vision-transformer (ViT) guide at $J_2/J_1 = 0.5$, with eight transformer layers, embedding dimension 60 and ten attention heads. In the controlled sign benchmarks we keep its modulus $a(x) = |\psi_{\text{ViT}}(x)|$ fixed and replace only its sign; the neural state is not retrained during that comparison.

The 8×8 fixed-node calculations use population $M = 128$ and projection length $\beta = 1.2$. Whenever only two independent fixed-node populations are available, we show both population means and use half their difference as a reproducibility scale. We do not interpret that two-population spread as a precision statistical error bar. Sampling and estimator details for each plotted quantity are retained with the source data in the accompanying reproducibility files.

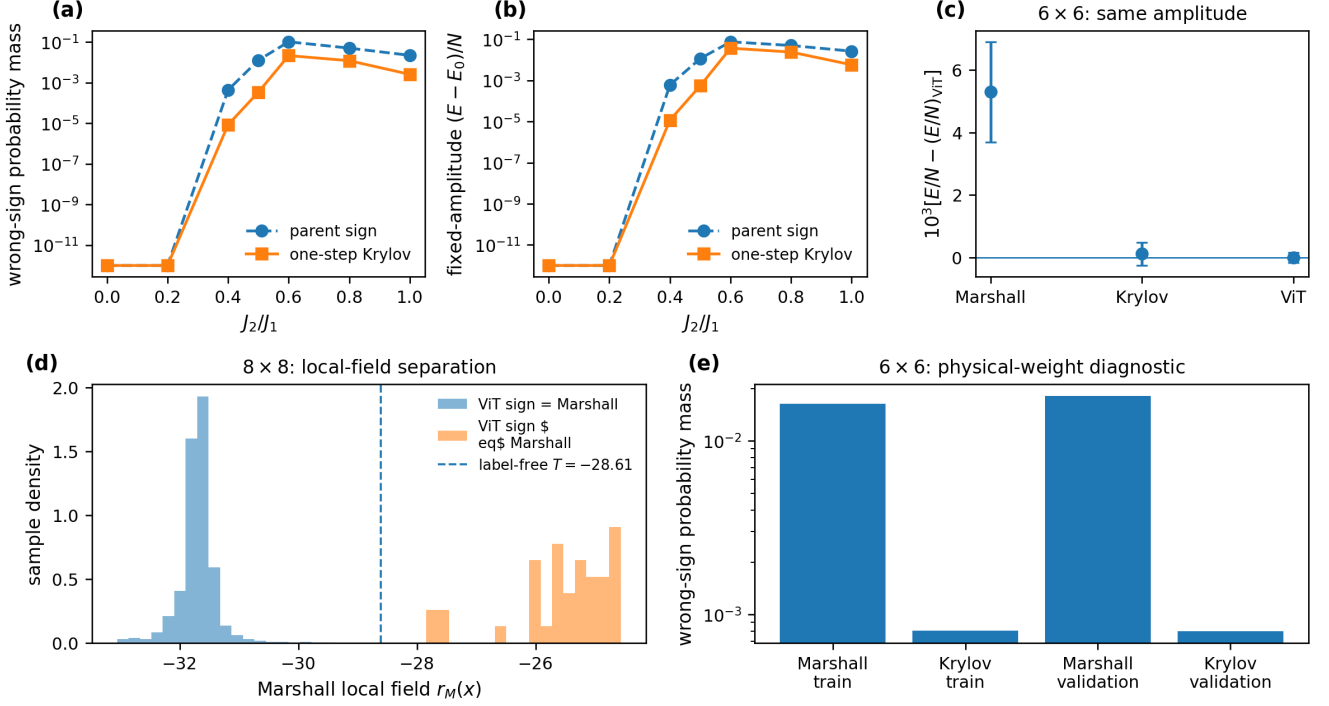


FIG. 2. Why one Krylov step is useful. (a) Exact 4×4 wrong-sign probability mass before and after the sign update. (b) The corresponding fixed-amplitude energy error. At strong frustration the parent state is a stripe sign pattern rather than Marshall. (c) 6×6 energies obtained with the identical ViT modulus and three different sign structures, shown relative to the original ViT signs. (d) On an 8×8 sample, the Marshall local field separates configurations whose stored ViT sign agrees or disagrees with Marshall; ViT signs are used only for this offline coloring, not to choose the threshold. (e) Independent 6×6 train and validation sets confirm an approximately twenty-fold reduction of physically weighted wrong-sign mass.

III. ONE KRYLOV STEP RECONSTRUCTS THE IMPORTANT SIGNS

We first isolate the sign problem by supplying a high-quality positive amplitude and changing only the signs. Figure 2(a,b) uses the exact 4×4 ground-state modulus. For $J_2/J_1 \leq 0.6$ the parent sign is the Marshall pattern; on the strongly stripe-dominated side we use the corresponding stripe reference. Thus the sweep tests reconstruction within the physically appropriate sign basin rather than asking a single Marshall reference to describe every coupling.

At the maximally frustrated point $J_2/J_1 = 0.5$, the Marshall sign overlap is $O_s = 0.974538$, corresponding to $w_s = 1.27 \times 10^{-2}$. A single update raises the overlap to 0.999319 and leaves only $w_s = 3.41 \times 10^{-4}$. The same change reduces the fixed-amplitude energy error from 1.16×10^{-2} to 5.51×10^{-4} per site. Thus the improvement is not merely a better classification score: at fixed exact amplitude it removes almost all of the energy penalty associated with the wrong signs.

The 6×6 panel provides a less oracle-dependent test. We take the positive amplitude from a trained ViT neural quantum state and evaluate three signed states with that

same amplitude. Their energies per site are

$$e_{\text{Marshall}} = -0.4981534, \quad (12)$$

$$e_{\text{Krylov}} = -0.5033337, \quad (13)$$

$$e_{\text{ViT}} = -0.5034504. \quad (14)$$

The paired difference between the Krylov and ViT signs is $e_{\text{Krylov}} - e_{\text{ViT}} = 1.17(287) \times 10^{-4}$, whereas the Krylov signs improve over Marshall by 4.79×10^{-3} per site. Because the amplitude is identical in all three evaluations, this comparison isolates the physical quality of the sign structure.

A. Why does one step work so well?

The local quantity in Eq. (4) can be decomposed in the Marshall gauge as

$$r_M(x) = D(x) + F_{J_1}(x) + F_{J_2}(x), \quad (15)$$

where D is diagonal, F_{J_1} is the unfrustrated nearest-neighbor hopping contribution, and F_{J_2} is the frustrating next-nearest-neighbor contribution. The useful sign information is carried predominantly by the competition of the two off-diagonal terms, weighted by neighboring amplitudes, rather than by a classical diagonal motif.

Figure 2(d) makes this separation visible on 8×8 . Although the two distributions overlap, the sign-defect population is strongly biased toward one side of the scalar threshold. On the independent 6×6 mechanism audit, the physically weighted wrong-sign mass changes from 0.01635 to 8.08×10^{-4} on the training sample and from 0.01811 to 8.02×10^{-4} on validation. The first Krylov step therefore removes about 95% of the relevant sign-defect weight.

This success does not imply that repeated bare Krylov steps form a convergent sign solver. The first Krylov vector also multiplies the amplitude by $|T - r_s(x)|$. If that distorted amplitude is retained, a second unsupervised threshold acts on a residual defect population that is already extremely small and is dominated by amplitude and local-energy outliers. Direct two-step tests on 6×6 worsen the fixed-amplitude energy. This is why our feedback loop returns to a fixed-node amplitude solve before asking for another sign update; quantitative two-step data are summarized in Appendix A.

IV. EXACT CLOSURE OF THE FIXED-NODE-KRYLOV LOOP

The previous section assumed that a useful positive amplitude was available. We next ask whether the two operations in Fig. 1 can generate each other's input. This can be tested without sampling noise on 4×4 . We target $J_2/J_1 = 0.5$ but initialize the guide with the exact *nonfrustrated* ($J_2 = 0$) ground-state modulus and Marshall signs. At each iteration we solve the lattice fixed-node problem exactly for the current guide, use its positive modulus as the next amplitude, and then recompute the one-step sign rule of Eq. (6). The target frustrated ground state is used only to measure the errors plotted below.

The result is shown in Fig. 3. The initial amplitude fidelity to the target state is only 0.7132, while the initial sign overlap is the Marshall value 0.974538. The fixed-node amplitude improves rapidly; sign changes then occur in discrete bursts when the updated amplitude moves the local-energy threshold across a new set of configurations. After two iterations the sign overlap is already 0.995945, after twenty iterations it is 0.9999982, and by iteration 100 the exact target sign pattern is reached. At that point the amplitude fidelity is 0.9999946 and the guide energy lies only 2.60×10^{-5} above the exact ground-state energy.

The staircase structure in panel (a) is useful conceptually. The sign update does not need to improve at every iteration. Fixed-node projection can first reshape the positive amplitude while the sign pattern remains unchanged; once enough amplitude information has accumulated, Eq. (6) crosses another sign boundary. The exact loop therefore demonstrates genuine feedback between the two sectors rather than repeated application of the same sign rule to a frozen amplitude.

V. SCALING THE SIGN UPDATE TO 8×8

The exact loop is a proof of principle, not a scalable implementation, because a generic amplitude cannot be tabulated on large lattices. Before addressing that hand-off, we first test whether the sign update itself remains useful when the system size is increased.

On 8×8 we start from the same trained ViT amplitude used to construct the one-step sign guide and perform lattice fixed-node projection with population $M = 128$. Two independent Krylov-sign fixed-node runs give total energies -31.88702 and -31.88383 , while matched Marshall-sign controls give -31.70383 and -31.65113 . The mean difference is

$$E_{\text{Krylov-FN}} - E_{\text{Marshall-FN}} = -0.20795, \quad (16)$$

with a two-replica spread proxy of 0.02640. We use this spread only as a reproducibility diagnostic; with two populations it is not a precision asymptotic uncertainty estimate.

Figure 4 places this result next to published calculations on the identical 8×8 periodic geometry. The Krylov-sign fixed-node mean is

$$E/N = -0.498210, \quad (17)$$

compared with -0.494961 for the Marshall-sign fixed-node control. The Krylov-guided value lies in the same narrow energy range as the high-bond dimension DMRG/FAMPS and RBM+pair-product benchmarks compiled in Ref. [8]. We do not interpret the small ordering among these numbers as a precision ranking: our fixed-node result has finite-population systematics and only two independent populations. The useful conclusion is more limited and more robust: the one-step reconstructed sign structure survives to 8×8 and removes most of the large energy penalty of the Marshall signs.

At this point the bottleneck changes. Fixed-node projection can be sampled at 8×8 , and the one-step sign rule can be evaluated from a callable guide. What is missing is a scalable map from a cloud of fixed-node walkers back to a callable positive amplitude. A low-dimensional histogram correction works as a diagnostic on smaller lattices, but nested evaluation of such corrections becomes prohibitively expensive at 8×8 . We therefore turn to direct learning of the amplitude update in Eq. (11).

VI. LEARNING THE FIXED-NODE AMPLITUDE FROM WALKERS

Equation (11) provides an exact target for the learning problem. Before using a neural network, we can apply that geometric-mean update exactly on 4×4 . Figure 5(a) shows two initial conditions. Starting from the exact target modulus but Marshall signs, the first ideal maximum-likelihood half step followed by the Krylov sign update changes the energy from -8.27291 to -8.44579 ,

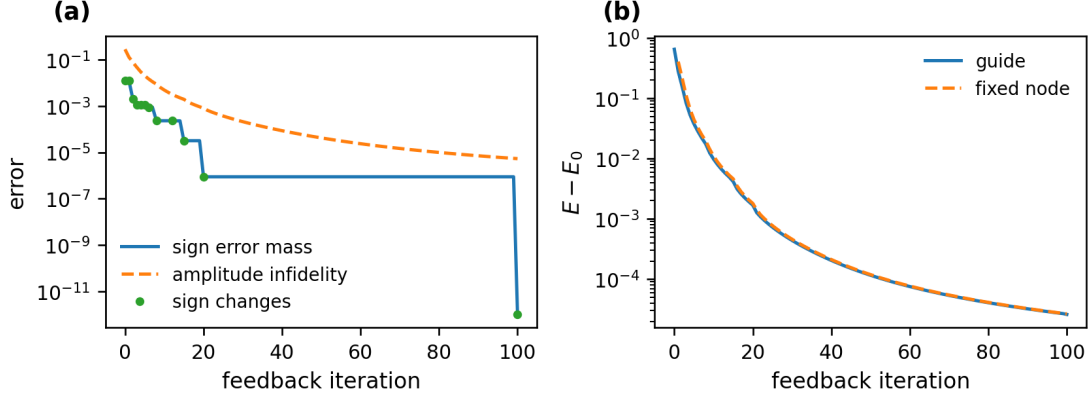


FIG. 3. Exact 4×4 feedback loop at $J_2/J_1 = 0.5$. (a) Exact sign-error mass and amplitude infidelity versus iteration. Markers on the sign-error curve indicate iterations at which the sign pattern actually changes. (b) Guide and fixed-node energy errors relative to the exact ground-state energy. The update itself never uses the target ground-state wave function.

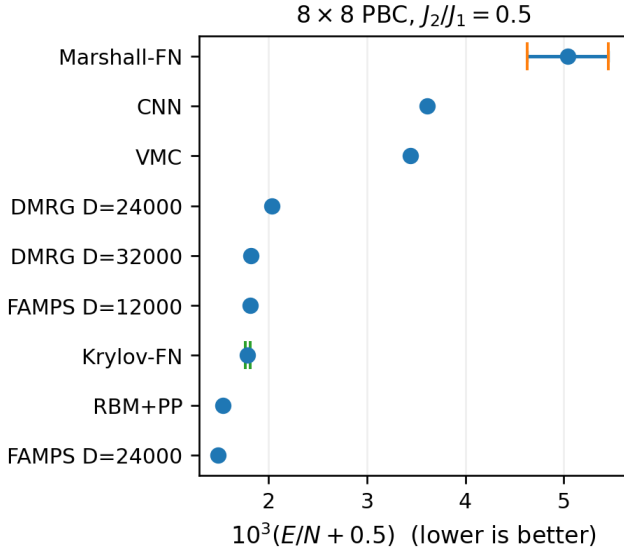


FIG. 4. Same-geometry energy benchmark for the 8×8 periodic J_1 - J_2 model at $J_2/J_1 = 0.5$. We plot $10^3(E/N + 0.5)$ so that lower values correspond to lower energies. Literature numbers are taken from Table I of Ref. [8]. Vertical ticks on our two fixed-node entries show the individual populations; their horizontal error bars are two-replica spread proxies.

close to the exact $E_0 = -8.45792$. Starting instead from a uniform modulus is deliberately much harder: the first update is nonmonotone, but subsequent fixed-node/half-step iterations recover and approach the correct low-energy basin. Thus the geometric-mean target is locally contracting toward each current fixed-node amplitude, but it does not define a global energy-descent theorem.

The finite-sample problem is more subtle. We fine-tune the existing ViT amplitude rather than replacing it with a newly initialized network. Let g_1 and g_2 denote maximum-likelihood gradients estimated from two inde-

pendent $M = 128$ fixed-node populations. At the initial 8×8 guide, their Euclidean cosine is

$$\frac{g_1 \cdot g_2}{\|g_1\| \|g_2\|} = -0.853. \quad (18)$$

A naive Euclidean update therefore points in a direction that does not generalize even at infinitesimal step size.

We instead use stochastic reconfiguration (SR) [9]. With empirical Fisher matrix S and diagonal shift λ , its update is

$$\delta\theta = (S + \lambda I)^{-1} g_1. \quad (19)$$

Using $\lambda = 1$ and a matrix-free solve, the same two populations give the held-out directional derivative

$$g_2^T \delta\theta = +3.85, \quad (20)$$

despite the negative Euclidean cosine. The finite line search in Fig. 5(b) confirms that this is not merely an infinitesimal effect: at $\eta = 0.01$ the held-out log-likelihood gain is $+0.0156$, while the importance-sampling effective sample size remains 3236 out of 4096.

This is the scalable amplitude result established here. It shows that fixed-node walker information can be converted into a neural-amplitude direction that transfers between independent populations. It does *not* yet establish convergence of a repeatedly learned 8×8 fixed-node-Krylov loop. In particular, likelihood improvement is a surrogate for the amplitude handoff, whereas the projected fixed-node energy remains the physical acceptance criterion for future iterations.

VII. DISCUSSION

The numerical picture that emerges is simpler than trying to learn an entire frustrated wave function in one operation. Once a useful positive amplitude is available,

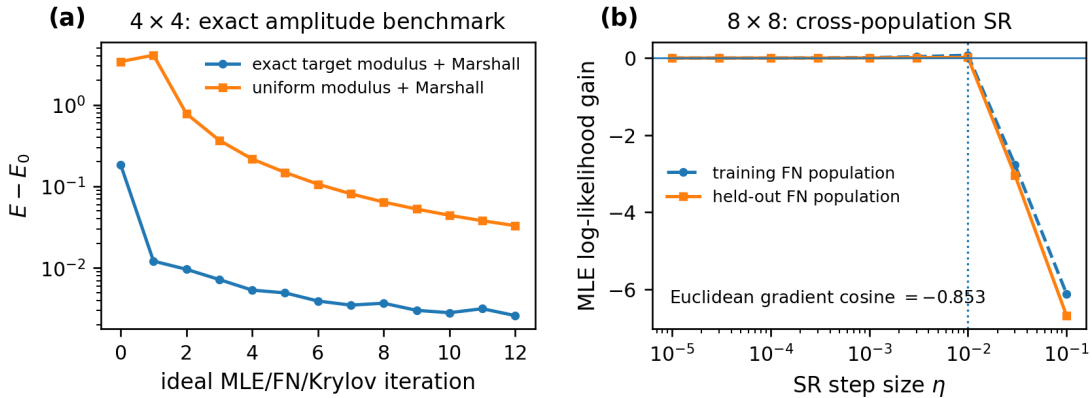


FIG. 5. Amplitude-learning benchmarks. (a) Exact 4×4 iterations of the ideal walker-MLE half step, Eq. (11), interleaved with the Krylov sign update. The uniform-amplitude start illustrates that the first step need not lower the energy globally. (b) On 8×8 , a stochastic-reconfiguration (SR) direction is trained on one fixed-node walker population and evaluated on an independent population. The held-out log-likelihood remains positive up to a finite step and is maximal near $\eta = 0.01$.

the dominant correction to the sign structure is already encoded in the amplitude-weighted local action of the Hamiltonian. A single shifted Krylov vector converts that information into a scalar threshold problem. Exact diagonalization shows that this removes almost all of the sign-defect weight at $J_2/J_1 = 0.5$, and the same-amplitude 6×6 test shows that the reconstructed signs have essentially the same energetic effect as the signs of the trained neural state.

Fixed-node projection plays a complementary role. It does not infer a new sign pattern: by construction it keeps the current signs fixed and solves a sign-free positive-amplitude problem. The sign change occurs only when the resulting amplitude is inserted back into Eq. (6). The exact 4×4 calculation demonstrates that this composition can cross many distinct sign patterns and eventually recover the exact frustrated ground state from a nonfrustrated starting point.

The scaling tests separate two questions that are often conflated. The Krylov sign rule itself remains useful through 8×8 , where its fixed-node energy is competitive with established same-geometry variational benchmarks. The difficult step is instead statistical: a fixed-node walker population represents the mixed distribution of Eq. (10), not a callable wave-function modulus. Direct Euclidean maximum-likelihood gradients are population dependent at 8×8 . A natural-gradient/SR metric repairs that first transfer test, but a robust repeated large-system feedback loop still requires a physical acceptance test and controlled population statistics.

Three boundaries of the present result are therefore important. First, the one-step rule is not a generic solution of the sign problem: it requires a useful starting amplitude and a parent sign basin. Second, the excellent first step should not be extrapolated to an undamped Krylov sequence; the direct second step is demonstrably worse. Third, the exact small-system closure does not prove polynomial scaling of the learned amplitude

handoff. What it does establish is a concrete decomposition of the problem in which sign reconstruction is inexpensive and quantitatively accurate, while the remaining hard object—a transferable positive amplitude learned from projector data—can be isolated and tested independently.

VIII. CONCLUSION

For the frustrated square-lattice J_1 - J_2 model, accurate amplitudes contain enough local Hamiltonian information to reconstruct most of the non-Marshall sign structure with one projected Krylov step. Exact diagonalization, same-amplitude neural-state tests, and 8×8 fixed-node calculations all support this statement. Fixed-node projection can in turn improve the amplitude without a sign problem, and on 4×4 the two updates form a self-correcting loop that reaches the exact signs.

For large systems the central unresolved issue is no longer how to propose a sign change, but how to turn sampled fixed-node amplitude information into a stable callable guide. The mixed-distribution identity $a_{\text{new}} \propto \sqrt{a\phi_{\text{FN}}}$ gives the correct learning target, and stochastic reconfiguration produces a direction that transfers between independent 8×8 walker populations. Closing that learned loop repeatedly, with fixed-node energy rather than likelihood as the acceptance criterion, is the remaining step.

Appendix A: Why a direct second Krylov step is not used

A natural alternative to the feedback construction would be to retain the full first Krylov vector and apply a second Krylov step. We tested this directly on 6×6 . The first operation changes not only the signs but also

the modulus,

$$a_1(x) = |T_1 - r_0(x)|a_0(x), \quad (\text{A1})$$

whose multiplier has a coefficient of variation of approximately 0.18 on the physical diagnostic sample.

Across three unsupervised second-step thresholds, the physical sign-change mass is only 0.8–2.0%, but the fixed-amplitude energy rises by 0.65–0.96 in total energy relative to the one-step signs. The residual correctable sign defects are much rarer than the population selected by an ordinary second threshold. Restoring the original high-quality amplitude removes most of the directly observed bad second-step flips. We therefore interpret the failure as amplitude distortion and rare-tail sensitivity, not as evidence that the dominant first-step sign correction was spurious.

Appendix B: Statistical conventions

Exact-diagonalization quantities are deterministic up to numerical eigensolver precision. For the 6×6 fixed-amplitude comparison we use independent chains and report a correlation-robust between-chain standard error. For 8×8 fixed-node calculations the manuscript reports two independent populations. The quantity shown as an error bar is the between-population spread divided by two, equal to the usual standard-error formula for two samples. Because two populations are insufficient to establish asymptotic normal statistics, we refer to this quantity throughout as a *two-replica spread proxy*, not as a precision statistical uncertainty.

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